

G R DIKSHITH

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SUMMARY

Software Engineer specializing in Artificial Intelligence and Machine Learning, with experience building scalable, production-ready AI systems and backend architectures. Developed an AI-powered bid document extractor that reduced manual processing by 85%, a heart failure prediction model achieving 87.5% accuracy (91.1% cross-validation), and a multi-agent AI cyber defense framework for real-time anomaly detection. Skilled in Python, SQL, FastAPI, TensorFlow, PyTorch, LLM integration, Docker, and cloud deployment, with a strong foundation in data structures, algorithms, and system design. I also publish technical blogs on AI and data engineering, reflecting a focus on practical implementation and scalable software solutions.

EXPERIENCE

Orfiq AI solutions Ltd – AI Engineer Jan 2026 - Present (London-United Kingdom -remote)

- Architect and deploy production-grade Generative AI systems using Azure OpenAI and multi-agent LLM workflows.
- Design scalable FastAPI microservices for AI inference, RAG/GraphRAG (Neo4j + Vector DB), and real-time APIs.
- Built autonomous 16+ agents architectures with structured orchestration for complex reasoning tasks.
- Developed scalable FastAPI-based AI services integrated with Azure OpenAI.

Hardee Flexipach – Web & Backend Development Intern

Aug 2024–Oct 2024

- Contributed to the development and maintenance of internal company websites.
- Implemented backend logic and functionality using Python, Java, and C.
- Collaborated with the tech team to improve system performance and code quality.
- Gained practical experience in SDLC, debugging, and team-based development.

GitHub – Open Source Contributor

Bangalore, India 2023–2026

- Delivered 15+ production-grade PRs to enterprise-scale open-source ecosystems (Apache Airflow, Matplotlib, Prefect).
- Enhanced scheduler logic, visualization modules, and workflow orchestration components improving execution efficiency by 12–18%.
- Implemented robust unit/integration tests and CI optimizations improving release stability.
- Participated in cross-timezone code reviews and design discussions for scalable open-source architecture.

SKILLS

- **Programming & Tools:** Python, SQL, REST APIs, Flask, FastAPI, Git, basic Docker, API integration
- **Data Analytics & ML:** Random Forest, XGBoost, Feature Engineering, Model Evaluation, LangChain, CI/CD pipelines
- **Deep Learning & AI:** TensorFlow, PyTorch, CNNs, RNNs, Transformers, NLP, LangChain, GPT-4o-mini, HuggingFace
- **Data Visualization:** Matplotlib, Seaborn, Tableau, Power BI
- **Soft Skills:** Problem Solving, Analytical Thinking, Teamwork, Communication, Adaptability

EDUCATION

M.S. Ramaiah University of Applied Sciences, Bangalore

2022–2026

Bachelor of Technology in Computer Science and Engineering

RESEARCH PUBLICATIONS / PROJECTS

Carbonex: AI-Governed Carbon Credit Exchange – Digital Carbon Passport & Trust-Based Trading 2026

TypeScript • Next.js • React • Node.js • SHA-256 Cryptography • Tailwind CSS • Recharts

- Abstract Accepted at the **4th International Conference on Advancing Sustainable Futures (ICASF 2027)**, Abu Dhabi University, UAE.
- Impact: Developed an immutable, high-speed carbon trading ecosystem that eliminates double-counting of credits and reduces market manipulation through dynamic AI pricing and cryptographic verification.
- Engineered a Secure Hash-Ledger Verification system utilizing SHA-256 and the STAVP protocol to immutably track carbon offset issuance, transfers, and retirements with guaranteed data provenance.
- Built an AI-driven dynamic pricing and Trust Score engine that computes optimal credit valuations based on real-time market supply, demand, volatility, and corporate sustainability metrics.

- Designed a scalable n-tier architecture featuring a responsive Next.js frontend, interactive Recharts dashboards, secure Role-Based Access Control (RBAC) via JWT, and Next.js serverless API routes.

Multi-Agent AI Defense Against AI-Powered Attacks

Research Project (2025)

Python • Flask • Scikit-learn • MongoDB

- Accepted for Oral Presentation at the International Conference on **Integrated Intelligence and Cognitive Engineering (ICIICE 2026)**, Dubai, UAE.
- **Author and Corresponding Author of the book chapter** “A HoneyBee-Inspired Metaheuristic Multi-Agent Cyber Defense Framework for Adaptive Deception and Autonomous Response” accepted for publication in the edited volume *Metaheuristic Optimization for Social Good*.
- Proposed and designed a HoneyBee-inspired multi-agent cyber defense architecture integrating intelligent threat detection, adaptive deception, autonomous response, and continuous learning.
- Developed the Detect–Mislead–Neutralize–Learn security lifecycle leveraging XGBoost, CNN-LSTM, Deep Q-Network (DQN) reinforcement learning, adaptive honeypots, and distributed threat intelligence sharing.
- Designed a scalable agent-based cybersecurity framework utilizing Apache Kafka, Flask-based deception environments, machine learning, deep learning, and reinforcement learning for autonomous cyber defense operations.
- Evaluated the framework using CIC-IDS2017 and simulated enterprise attack scenarios, achieving 98.24% detection accuracy, 98.51% precision, 98.10% recall, 0.992 ROC-AUC, and a 1.85% false positive rate.
- Achieved a deception engagement rate of 92.40% through adaptive attacker redirection and intelligence collection

Bid Document Extractor – AI-Powered Intelligent Document Processing System

(2025)

Python • LangChain • OpenRouter API • GPT-4o-mini • Regex • JSON • PDF/HTML Parsing

- Impact: Reduced manual bid parsing time by 85% and improved structured data accuracy to **90%**, demonstrating strong LLM integration and automation design skills.
- Built an AI-powered bid document extractor using GPT-4o-mini and LangChain with hybrid regex logic, achieving 92% accuracy.
- Automated end-to-end data extraction and JSON generation for seamless ERP and Bid Management integration.

Heart Failure Prediction Model – Logistic Regression

Jun 2025

Python • Scikit-learn • Pandas • EDA • Feature Engineering • Logistic Regression

- Impact: **Achieved 87.5% accuracy, 91.11% CV score, and 87.43% ROC-AUC**, showcasing strong ML modeling and evaluation skills.
- Built a Logistic Regression model with complete preprocessing, feature engineering, and visualization workflows.
- Performed EDA, encoding, scaling, and multi-metric evaluation for reliable heart disease prediction.

CERTIFICATIONS

- IBM Generative AI for Software Developers – Coursera (2025)
- OpenGL – Infosys Springboard (2024)
- IBM AI Engineering Professional Certificate – Coursera (2025)
- Cloud Computing – NPTEL (2024)
- Machine Learning with Python – Infosys Springboard (2024)
- Neural networks and deep learning - (2025)